

Table 1. English TruthfulQA results with evaluated-model KV cache disabled. Full-protocol rows report mean $\pm$ standard error across five decoding repetitions; refreshed Llama-3-8B H_∞ , GPT-2 XL, OLMo-2-32B, and compact Qwen-2.5-32B rows report five-group delete-one-group question-jackknife standard errors.

	Method	True (%) $\uparrow$	Informative (%) $\uparrow$	T $\times$ I (%) $\uparrow$	Instruction relevance (0–2) $\uparrow$	Fluency (0–2) $\uparrow$
Gemma-2-2B	Original	49.79 $\pm$ 0.22	95.35 $\pm$ 0.22	47.48 $\pm$ 0.24	1.35 $\pm$ 0.02	1.23 $\pm$ 0.01
	ITI	50.65 $\pm$ 0.39	95.01 $\pm$ 0.13	48.12 $\pm$ 0.38	1.39 $\pm$ 0.02	1.29 $\pm$ 0.01
	ActAdd	60.37 $\pm$ 0.66	86.49 $\pm$ 0.61	52.21 $\pm$ 0.68	1.22 $\pm$ 0.01	1.08 $\pm$ 0.01
	Mean-AcT	48.00 $\pm$ 0.51	93.76 $\pm$ 0.34	45.01 $\pm$ 0.50	1.39 $\pm$ 0.02	<u>1.29 $\pm$ 0.01</u>
	Linear-AcT	46.71 $\pm$ 0.76	94.15 $\pm$ 0.12	43.97 $\pm$ 0.72	<u>1.41 $\pm$ 0.01</u>	1.28 $\pm$ 0.01
	PID-AcT	47.07 $\pm$ 0.52	93.83 $\pm$ 0.34	44.17 $\pm$ 0.51	1.38 $\pm$ 0.01	1.29 $\pm$ 0.01
	ODESteer	56.18 $\pm$ 0.29	<u>95.76 $\pm$ 0.26</u>	53.80 $\pm$ 0.32	1.37 $\pm$ 0.02	1.23 $\pm$ 0.01
	S-PID	52.51 $\pm$ 0.44	94.59 $\pm$ 0.30	49.67 $\pm$ 0.45	1.39 $\pm$ 0.01	1.22 $\pm$ 0.00
	A-LQR	62.33 $\pm$ 0.25	91.65 $\pm$ 0.39	<u>57.12 $\pm$ 0.34</u>	1.29 $\pm$ 0.02	1.15 $\pm$ 0.01
	H$_\infty$ (ours)	70.11 $\pm$ 0.70	79.34 $\pm$ 0.50	55.62 $\pm$ 0.66	1.14 $\pm$ 0.01	1.07 $\pm$ 0.01
Llama-3-8B	Original	49.28 $\pm$ 0.98	96.99 $\pm$ 0.29	47.79 $\pm$ 0.96	1.33 $\pm$ 0.03	1.30 $\pm$ 0.01
	ITI	57.14 $\pm$ 0.79	97.53 $\pm$ 0.09	55.72 $\pm$ 0.77	1.38 $\pm$ 0.02	1.30 $\pm$ 0.01
	ActAdd	56.72 $\pm$ 0.99	88.69 $\pm$ 0.82	50.30 $\pm$ 0.99	1.23 $\pm$ 0.02	1.17 $\pm$ 0.01
	Mean-AcT	47.47 $\pm$ 0.71	97.55 $\pm$ 0.26	46.30 $\pm$ 0.70	1.36 $\pm$ 0.01	1.31 $\pm$ 0.01
	Linear-AcT	48.20 $\pm$ 0.69	97.50 $\pm$ 0.19	47.00 $\pm$ 0.68	1.35 $\pm$ 0.02	<u>1.33 $\pm$ 0.00</u>
	PID-AcT	47.69 $\pm$ 0.21	97.60 $\pm$ 0.26	46.54 $\pm$ 0.24	1.34 $\pm$ 0.01	1.31 $\pm$ 0.01
	ODESteer	<u>95.81 $\pm$ 0.06</u>	9.28 $\pm$ 0.36	8.89 $\pm$ 0.35	0.01 $\pm$ 0.00	0.03 $\pm$ 0.01
	S-PID	57.55 $\pm$ 0.21	97.80 $\pm$ 0.27	56.28 $\pm$ 0.26	1.41 $\pm$ 0.02	1.27 $\pm$ 0.01
	A-LQR	58.95 $\pm$ 0.27	<u>98.07 $\pm$ 0.20</u>	57.81 $\pm$ 0.29	<u>1.44 $\pm$ 0.03</u>	1.21 $\pm$ 0.01
	H$_\infty$ (ours)	76.50 $\pm$ 1.11	80.29 $\pm$ 0.59	61.42 $\pm$ 1.15	0.99 $\pm$ 0.02	1.00 $\pm$ 0.01
Qwen-2.5-14B	Original	55.89 $\pm$ 0.33	96.47 $\pm$ 0.15	53.92 $\pm$ 0.33	1.46 $\pm$ 0.01	1.24 $\pm$ 0.01
	ITI	61.62 $\pm$ 0.62	95.32 $\pm$ 0.35	58.73 $\pm$ 0.63	1.52 $\pm$ 0.01	1.25 $\pm$ 0.01
	ActAdd	64.04 $\pm$ 0.42	87.61 $\pm$ 0.61	56.11 $\pm$ 0.53	1.17 $\pm$ 0.01	0.99 $\pm$ 0.01
	Mean-AcT	55.35 $\pm$ 1.09	96.03 $\pm$ 0.21	53.15 $\pm$ 1.05	1.50 $\pm$ 0.01	1.24 $\pm$ 0.01
	Linear-AcT	56.28 $\pm$ 0.95	96.18 $\pm$ 0.28	54.13 $\pm$ 0.93	1.50 $\pm$ 0.01	1.23 $\pm$ 0.01
	PID-AcT	56.08 $\pm$ 0.89	96.28 $\pm$ 0.15	54.00 $\pm$ 0.86	1.49 $\pm$ 0.01	1.24 $\pm$ 0.01
	ODESteer	61.40 $\pm$ 0.60	94.00 $\pm$ 0.15	57.71 $\pm$ 0.57	1.41 $\pm$ 0.01	1.21 $\pm$ 0.00
	S-PID	66.93 $\pm$ 0.93	96.52 $\pm$ 0.22	64.60 $\pm$ 0.91	1.51 $\pm$ 0.01	<u>1.26 $\pm$ 0.01</u>
	A-LQR	71.82 $\pm$ 0.60	96.99 $\pm$ 0.28	69.66 $\pm$ 0.62	<u>1.53 $\pm$ 0.01</u>	<u>1.26 $\pm$ 0.01</u>
	H$_\infty$ (ours)	85.31 $\pm$ 0.33	89.40 $\pm$ 0.35	76.27 $\pm$ 0.42	1.34 $\pm$ 0.01	1.15 $\pm$ 0.00
Qwen-2.5-32B	Original	61.86 $\pm$ 2.08	98.53 $\pm$ 0.46	60.95 $\pm$ 2.07	1.49 $\pm$ 0.02	1.27 $\pm$ 0.02
	S-PID	73.84 $\pm$ 1.87	96.82 $\pm$ 0.50	71.49 $\pm$ 1.84	1.40 $\pm$ 0.04	<u>1.26 $\pm$ 0.02</u>
	A-LQR	78.73 $\pm$ 1.97	<u>97.31 $\pm$ 0.60</u>	<u>76.61 $\pm$ 1.97</u>	<u>1.53 $\pm$ 0.04</u>	1.23 $\pm$ 0.01
	H$_\infty$ (ours)	82.15 $\pm$ 2.10	92.42 $\pm$ 1.25	75.92 $\pm$ 2.19	1.42 $\pm$ 0.03	1.16 $\pm$ 0.01
OLMo-2-32B	Original	56.92 $\pm$ 2.36	98.78 $\pm$ 0.34	56.22 $\pm$ 2.15	1.59 $\pm$ 0.03	1.43 $\pm$ 0.02
	A-LQR	87.76 $\pm$ 1.50	93.15 $\pm$ 1.22	81.74 $\pm$ 0.65	1.31 $\pm$ 0.03	1.02 $\pm$ 0.01
	H$_\infty$ (ours)	92.66 $\pm$ 0.73	93.15 $\pm$ 0.80	86.31 $\pm$ 0.42	1.38 $\pm$ 0.03	1.12 $\pm$ 0.01
GPT-2 XL	Original	28.89 $\pm$ 1.56	94.25 $\pm$ 0.42	27.22 $\pm$ 1.35	0.82 $\pm$ 0.01	1.08 $\pm$ 0.01
	ITI	30.11 $\pm$ 0.73	95.84 $\pm$ 0.66	28.86 $\pm$ 0.58	0.90 $\pm$ 0.02	1.08 $\pm$ 0.01
	ActAdd	36.60 $\pm$ 2.10	88.86 $\pm$ 1.46	32.52 $\pm$ 1.50	0.78 $\pm$ 0.03	1.02 $\pm$ 0.02
	Mean-AcT	30.35 $\pm$ 1.37	95.84 $\pm$ 1.07	29.09 $\pm$ 1.40	<u>0.91 $\pm$ 0.01</u>	1.02 $\pm$ 0.02
	Linear-AcT	29.50 $\pm$ 1.28	96.33 $\pm$ 0.91	28.42 $\pm$ 1.12	0.87 $\pm$ 0.02	<u>1.11 $\pm$ 0.02</u>
	PID-AcT	29.87 $\pm$ 1.66	<u>96.57 $\pm$ 1.05</u>	28.84 $\pm$ 1.48	0.89 $\pm$ 0.03	1.09 $\pm$ 0.01
	ODESteer	37.58 $\pm$ 2.86	92.90 $\pm$ 0.64	34.91 $\pm$ 2.58	0.89 $\pm$ 0.03	1.06 $\pm$ 0.02
	S-PID	28.40 $\pm$ 1.16	86.90 $\pm$ 1.49	24.68 $\pm$ 1.00	0.84 $\pm$ 0.03	0.84 $\pm$ 0.02
	A-LQR	31.21 $\pm$ 1.98	92.04 $\pm$ 1.00	28.73 $\pm$ 1.85	0.84 $\pm$ 0.02	1.05 $\pm$ 0.02
	H$_\infty$ (ours)	46.51 $\pm$ 1.17	75.64 $\pm$ 1.74	35.18 $\pm$ 0.43	0.78 $\pm$ 0.03	0.76 $\pm$ 0.03